

MA'AN, Jordan

WMO: 403100

Lat: 30.167N

Lon: 35.783E

Elev: 3507

StdP: 12.93

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.1	33.6	12.0	11.5	51.8	16.0	14.0	54.5	42.2	41.3	34.9	39.9	3.2	280	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	97.9	67.7	95.7	66.9	93.5	65.8	71.5	92.3	69.9	90.0	68.5	87.9	6.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	108.2	77.7	64.0	101.7	78.1	62.2	95.3	77.5	37.7	91.8	36.1	89.8	35.0	87.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	23.1	19.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	46.0	49.4	56.7	62.9	72.1	77.4	80.4	80.6	76.0	68.5	57.1	49.0
	DBStd	13.56	5.70	6.07	7.23	7.58	6.14	4.42	4.29	3.86	4.18	5.83	5.51	6.30
	HDD50	353	154	76	20	3	0	0	0	0	0	0	7	93
	HDD65	2214	590	437	273	129	12	0	0	0	0	32	244	497
	CDD50	5739	29	58	228	390	685	822	942	948	781	574	220	62
	CDD65	2123	0	0	16	65	231	372	477	483	331	141	6	1
	CDH74	24128	0	9	193	748	2599	4336	5780	5673	3400	1317	61	12
	CDH80	11423	0	0	43	212	1062	2126	3019	3014	1526	416	4	1
(14) Wind	WSAvg	6.1	6.5	6.8	8.4	8.2	7.3	6.4	5.9	5.6	5.2	4.3	4.5	4.8
(15) Precipitation	PrecAvg	1.6	0.4	0.3	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.2	0.1	0.2
	PrecMax	3.7	1.8	1.8	1.1	1.1	1.0	0.2	0.0	0.0	0.2	2.1	0.9	1.6
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.4	0.4	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.4	0.2	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.4	75.5	85.4	90.7	96.6	97.9	101.6	100.7	97.5	91.0	80.0	74.7
		MCWB	50.2	53.7	56.7	61.0	64.8	65.4	70.1	71.3	65.1	62.0	57.4	53.2
	2%	DB	65.2	70.1	80.3	86.0	92.4	95.3	98.0	97.5	93.3	87.7	75.3	69.3
		MCWB	48.6	49.8	55.0	57.9	62.8	65.4	68.6	68.6	64.5	60.7	55.2	50.7
	5%	DB	61.4	65.4	75.2	82.0	89.4	93.2	95.3	95.2	90.7	85.0	71.8	64.4
		MCWB	47.3	47.9	52.8	56.6	62.0	64.3	67.5	67.8	63.5	60.3	53.9	50.0
	10%	DB	57.8	61.7	71.1	78.3	86.4	90.9	92.5	92.7	88.3	81.7	68.2	60.1
		MCWB	45.6	46.7	51.3	55.3	60.9	63.0	66.2	66.3	62.9	59.6	52.8	47.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	54.9	59.7	64.8	67.9	70.7	73.3	74.0	70.6	66.4	61.7	56.8
		MCDB	63.5	67.0	77.3	85.0	88.0	92.4	92.7	94.8	89.3	81.6	67.5	67.9
	2%	WB	50.7	51.9	56.6	60.8	65.6	68.6	70.9	72.1	68.0	64.2	58.2	53.4
		MCDB	61.7	66.4	75.5	79.1	87.6	91.0	92.9	92.4	85.2	79.9	69.8	64.6
	5%	WB	48.3	49.7	54.5	58.2	64.0	66.8	69.3	70.4	66.4	62.6	56.4	50.9
		MCDB	59.5	61.9	72.9	78.5	85.7	88.7	90.2	90.0	83.3	77.5	66.6	62.3
	10%	WB	46.1	47.8	52.2	56.3	62.1	64.9	68.0	69.1	65.0	61.1	54.8	48.7
		MCDB	56.0	59.8	69.1	76.0	83.6	86.3	88.6	87.6	82.2	76.9	64.9	58.3
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	19.9	21.9	23.1	24.9	26.0	24.8	26.0	25.4	23.6	20.8	19.8
		MCDBR	23.8	26.2	26.7	27.1	26.9	28.1	26.7	27.1	26.9	26.3	25.4	25.2
		MCWBR	13.5	13.0	12.4	12.8	11.4	11.0	11.5	11.0	11.0	10.8	12.5	13.5
	5% WB	MCDBR	22.3	22.9	24.3	25.2	25.7	27.6	25.5	26.6	25.9	24.0	21.3	22.2
		MCWBR	13.7	13.0	12.7	13.2	12.5	12.3	11.8	11.4	11.8	11.0	11.7	13.2
(40) Clear-Sky Solar Irradiance	taub		0.306	0.337	0.386	0.429	0.431	0.363	0.365	0.375	0.370	0.380	0.329	0.302
	taud		2.368	2.258	2.081	1.971	1.998	2.235	2.244	2.225	2.232	2.189	2.332	2.396
	Ebn at Noon		294	293	284	275	274	292	290	287	285	273	282	289
	Edh at Noon		32	39	50	58	57	45	44	44	43	42	33	30
(44) All-Sky Solar Radiation	RadAvg		1161	1463	1887	2207	2461	2683	2585	2388	2082	1650	1289	1093
	RadStd		50	62	68	52	91	33	48	39	56	58	49	41

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.79	N/A	-2.97	N/A	N/A	N/A	N/A	N/A	N/A	+272	+189

Nomenclature: See separate page